Differentiation Rules: Power, Product, Quotient, Chain

ANSWER KEY

Section 1: Power Rule and Basic Operations

1 Find the derivative of $f(x) = 4x^5 - 3x^3 + 7x - 9$.

a. $20x^4 - 9x^2 + 7$ ✓

b. $20x^4 - 9x^2$

c. $4x^4 - 3x^2 + 7$

d. $20x^4 - 6x^2 + 7$

By the power rule: $\frac{d}{dx}(4x^5) = 20x^4$, $\frac{d}{dx}(-3x^3) = -9x^2$, $\frac{d}{dx}(7x) = 7$, $\frac{d}{dx}(-9) = 0$.

2 Differentiate $g(x) = \frac{4}{\sqrt{x}} + \frac{3}{x^2}$.

a. $-2x^{-3/2} - 6x^{-3}$ ✓

b. $-2x^{-1/2} - 6x^{-3}$

c. $2x^{-3/2} - 6x^{-3}$

d. $-4x^{-3/2} - 3x^{-3}$

Rewrite as $4x^{-1/2} + 3x^{-2}$. $g'(x) = 4(-1/2)x^{-3/2} + 3(-2)x^{-3} = -2x^{-3/2} - 6x^{-3}$.

3 Find all points where the curve $y = x^3 - 3x^2 - 9x + 5$ has a horizontal tangent.

a. $x = -1$ **and** $x = 3$ ✓

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

$y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1) = 0 \Rightarrow x = 3$ or $x = -1$.

Section 2: Product and Quotient Rules

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4 Differentiate $h(x) = (x^2 + 3)(2x - 5)$.

a. $6x^2 - 10x + 6$ ✓

b. $6x^2 - 10x$

c. $4x^2 - 10x + 6$

d. $6x^2 + 6$

Product rule: $h'(x) = (2x)(2x - 5) + (x^2 + 3)(2) = 4x^2 - 10x + 2x^2 + 6 = 6x^2 - 10x + 6$.

5 Find $\frac{dy}{dx}$ for $y = \frac{3x - 1}{2x + 5}$.

a. $\frac{17}{(2x + 5)^2}$ ✓

b. $\frac{1}{(2x + 5)^2}$

c. $\frac{6x + 17}{(2x + 5)^2}$

d. $\frac{3}{2}$

Quotient rule: $\frac{3(2x + 5) - (3x - 1)(2)}{(2x + 5)^2} = \frac{6x + 15 - 6x + 2}{(2x + 5)^2} = \frac{17}{(2x + 5)^2}$.

6 If $f(2) = 3, f'(2) = -1, g(2) = 4, g'(2) = 2$, find $\left(\frac{f}{g}\right)'(2)$.

a. $-10/16 = -5/8$ ✓

b. $2/16 = 1/8$

c. $-2/16$

d. $-1/2$

$\frac{f'(2)g(2) - f(2)g'(2)}{[g(2)]^2} = \frac{(-1)(4) - (3)(2)}{4^2} = \frac{-4 - 6}{16} = -\frac{10}{16} = -\frac{5}{8}$.

Section 3: The Chain Rule

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7 Find $\frac{d}{dx}[(3x^2 - 5x + 2)^4]$.

a. $4(3x^2 - 5x + 2)^3(6x - 5)$ ✓

b. $4(3x^2 - 5x + 2)^3$

c. $12(6x - 5)^3$

d. $4(6x - 5)^3$

By chain rule: $\frac{d}{du}[u^4] \cdot u' = 4u^3 \cdot (6x - 5) = 4(3x^2 - 5x + 2)^3(6x - 5)$.

8 Find $\frac{dy}{dx}$ for $y = \sqrt{4 - x^2}$.

a. $\frac{-x}{\sqrt{4 - x^2}}$ ✓

b. $\frac{x}{\sqrt{4 - x^2}}$

c. $\frac{-1}{2\sqrt{4 - x^2}}$

d. $\frac{-2x}{\sqrt{4 - x^2}}$

$y = (4 - x^2)^{1/2} \Rightarrow y' = \frac{1}{2}(4 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{4 - x^2}}$.

9 If $h(x) = f(g(x))$ with $g(3) = 5$, $g'(3) = 4$, and $f'(5) = -2$, find $h'(3)$.

a. -8 ✓

b. 8

c. -10

d. 20

$h'(x) = f'(g(x))g'(x) \Rightarrow h'(3) = f'(g(3))g'(3) = f'(5)(4) = (-2)(4) = -8$.

Section 4: Trigonometric Derivatives with Chain Rule

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10 Differentiate $y = \sin(3x^2)$.

a. $6x\cos(3x^2)$ ✓

b. $3\cos(3x^2)$

c. $\cos(6x)$

d. $6x\sin(3x^2)$

Chain rule: $\frac{d}{dx}[\sin(u)] = \cos(u) \cdot u' = \cos(3x^2) \cdot 6x = 6x\cos(3x^2)$.

11 Find $\frac{d}{dx}[\tan^3(x)]$.

a. $3\tan^2(x)\sec^2(x)$ ✓

b. $3\tan^2(x)$

c. $3\sec^4(x)$

d. $\sec^6(x)$

Let $u = \tan x$: $\frac{d}{dx}[u^3] = 3u^2u' = 3\tan^2(x)\sec^2(x)$.

12 Find the second derivative y'' for $y = \cos(2x)$.

a. $-4\cos(2x)$ ✓

b. $4\cos(2x)$

c. $-2\sin(2x)$

d. $-4\sin(2x)$

$y' = -2\sin(2x)$. $y'' = \frac{d}{dx}[-2\sin(2x)] = -2(2\cos(2x)) = -4\cos(2x)$.

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